

BANARAS HINDU UNIVERSITY**B.A./B.Sc. (Hons) Semester IV Examination, 2025-26 (NEP)**

Subject: Mathematics

Paper No. MAT-MJ-41 : Calculus of Several Variables

Time: 2½ Hours

Full Marks: 70

REF NO. Conf/Even/N/2025-26/64

*Instructions: Answer **five** questions including Question 1 which is compulsory. The figures in the right-hand margin indicate marks.*

Question 1. (Compulsory)

- (a) Give the $\varepsilon - \delta$ definition of simultaneous limit of a real valued function in $\mathbb{R}^3$. [2]
- (b) Define Euclidean norm in $\mathbb{R}^n$. [2]
- (c) Give the definition of Gamma function. [2]
- (d) Show that the mixed partial derivatives of a real valued function in $\mathbb{R}^2$ need not be equal, with the help of an example. [3]
- (e) Give the definition of homogeneous function with an example. [3]
- (f) Define stationary point of a real valued function in $\mathbb{R}^2$. [2]
- (g) Define interior and boundary points of a set in the Euclidean space $\mathbb{R}^n$. [2]
- (h) Define triple integral in Cartesian coordinates. [2]
- (i) Give the statement of Liouville's Extension of Dirichlet's Theorem. [2]
- (j) Define Jacobian of an implicit function. [2]

Question 2.

- (a) If $x, y \in \mathbb{R}^n$ and $a \in \mathbb{R}$, then prove that—

$$\|x + y\| \leq \|x\| + \|y\|$$

$$\|ax\| = |a| \cdot \|x\|$$

[6]

- (b) State and prove mean value theorem for a function $f : D \subseteq \mathbb{R}^2 \rightarrow \mathbb{R}$. [6]

Question 3.

- (a) Investigate for simultaneous limit and continuity of the function

$$f(x, y) = \begin{cases} xy \frac{x^2 - y^2}{x^2 + y^2} & ; (x, y) \neq (0, 0), \\ 1 & ; (x, y) = (0, 0), \end{cases}$$

at $(0, 0)$. [6]

- (b) Prove that a function which is differentiable at a point admits of first-order partial derivatives at the point but the converse is not true. [6]

Question 4.

- (a) Find the equation of the tangent plane to the surface $2x^2y + y^2z = 3$ at the point $(-1, 1, 1)$. [6]
- (b) State and prove Young's theorem. [6]

Question 5.

- (a) Expand $f(y, x) = e^{xy}$ in power of $(x - 1)$ and $(y - 1)$ by Taylor's theorem as far as terms of third degree. [6]
- (b) Establish necessary and sufficient conditions for a function $f : D \subseteq \mathbb{R}^2 \rightarrow \mathbb{R}$ to have maximum or minimum value at a point $(a, b) \in D$. [6]

Question 6.

- (a) Find the volume under the plane $x + y + z = 6$ and above the triangle in the xy -plane bounded by $x = y$, $y = 0$ and $x = 3$. [6]
- (b) State and prove Euler's theorem for the function of n variables. [6]

Question 7.

- (a) By using Dirichlet's integral, find the volume of the solid bounded by $x = 0$, $y = 0$, $z = 0$ and the surface

$$\left(\frac{x}{a}\right)^{1/2} + \left(\frac{y}{b}\right)^{1/2} + \left(\frac{z}{c}\right)^{1/2} = 1$$

where $0 < c < b < a < \infty$. [6]

- (b) State and prove Legendre's duplication formula. [6]
